

## Unit 3 - Integral Calculus

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\* **Syllabus** - A Reduction Formulae, Beta and Gamma functions, Differentiation Under Integral Sign and Error functions.

\* **Marking Scheme** :-

SPPU QP Pattern (CM-IV)

Q.1 4 MCQ's for 8 marks and 2 MCQ's for 2 marks. = 10 Marks

Q.2(a) Reduction formulae - 5 marks

(b) Beta functions / Gamma functions - 5 marks

(c) DUIS / Error functions - 5 marks

} 15 marks

OR

Q.3(a) Beta function - 5 marks

(b) Gamma function / Beta function - 5 marks

(c) Error function / DUIS - 5 marks

} 15 marks

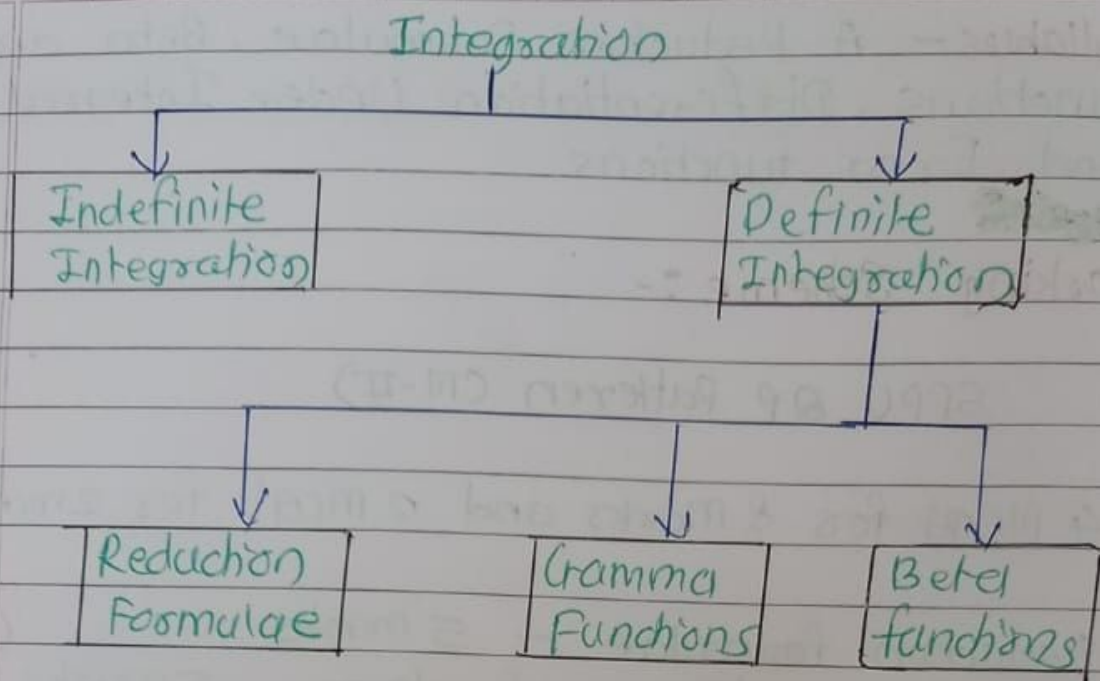
\* **Textbooks** :- Higher Engineering Mathematics by Dr.B.S. Grewal. (Khanna Publications).

chapter 6 - Page No - 239 to Page No 268

chapter 7 - Page No - 302 to Page No 315

\* **Reference book** - Advanced Engineering Mathematics by ERWIN KREYSZIG.

Page No 419 to Page No 426.



\* **Reduction Formulae**:- These are useful in evaluating integration of trigonometric functions with angular intervals (limits) which are not immediately solvable due to their higher powers. Ex)  $\int_0^{\pi/2} \sin^9(x) dx$ .

Without a reduction formula this would be difficult to integrate.

\* **Gamma Function**:- This is useful in evaluating integrals with higher powers of  $x$  multiplied by exponential functions with limits 0 to  $\infty$ .

Ex)  $\int_0^{\infty} e^{-x} x^{13} dx$

\* **Beta Functions**:- This is useful in evaluating integrals with higher powers of  $x$  multiplied by higher power of  $(1-x)$  with limits 0 to 1.

Ex)  $\int_0^1 x^9 (1-x)^{15} dx$



## \* Reduction Formulae \*

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### \* Formulae List:-

$$\textcircled{1} \int_0^{\pi/2} \sin^n(x) dx = \frac{(n-1)(n-3)(n-5) \dots 3 \cdot 1 \cdot \pi}{n(n-2)(n-4) \dots 4 \cdot 2 \cdot 2}$$

Where  $n$  is a positive even integer.

i.e.  $n = 2, 4, 6, 8, 10, \dots$

$$\textcircled{2} \int_0^{\pi/2} \cos^n(x) dx = \frac{(n-1)(n-3)(n-5) \dots 3 \cdot 1 \cdot \pi}{n(n-2)(n-4) \dots 4 \cdot 2 \cdot 2}$$

Where  $n$  is a positive even integer

i.e.  $n = 2, 4, 6, 8, 10, \dots$

$$\textcircled{3} \int_0^{\pi/2} \sin^n(x) dx = \frac{(n-1)(n-3)(n-5) \dots 4 \cdot 2 \cdot 1}{n(n-2)(n-4) \dots 5 \cdot 3 \cdot 1}$$

when  $n$  is a positive odd integer

i.e.  $n = 1, 3, 5, 7, 9, 11, \dots$

$$\textcircled{4} \int_0^{\pi/2} \cos^n(x) dx = \frac{(n-1)(n-3)(n-5) \dots 4 \cdot 2 \cdot 1}{n(n-2)(n-4) \dots 5 \cdot 3 \cdot 1}$$

IF  $n$  is odd positive integer

i.e.  $n = 1, 3, 5, 7, 9, 11, \dots$

$\therefore$  Reduction formulae =  $\frac{O}{A \cdot E} \cdot P$

where  $O$  = subtract odd numbers.

$A$  = As it is / Addition of powers

$E$  = subtract even numbers.

$P = \int_0^{\pi/2}$  when power is even

$= 1$  when power is odd.

Ex) Simplify  $\int_0^{\pi/2} \sin^6(x) dx$

$\Rightarrow$  Here  $n=6 \Rightarrow$  An even number.

$$\therefore \int_0^{\pi/2} \sin^6(x) dx = \frac{5}{\cancel{6}_2} \cdot \frac{\cancel{3}}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{5\pi}{32}$$

Ex) Simplify  $\int_0^{\pi/2} \cos^5(x) dx$

$\Rightarrow$  Here  $n=5 \Rightarrow$  An odd number

$$\therefore \int_0^{\pi/2} \cos^5(x) dx = \frac{4}{5} \cdot \frac{2}{3} \cdot 1 = \frac{8}{15}$$

\* Formulae:-

$$\int_0^{\pi/2} \sin^m(x) \cdot \cos^n(x) dx = \frac{[(m-1)(m-3)(m-5)\dots] \cdot [(n-1)(n-3)(n-5)\dots]}{(m+n)(m+n-2)(m+n-4)\dots}$$

Note: Do not consider 0 and negative numbers for calculations.

Ex)  $\int_0^{\pi/2} \sin^3 x \cos^4 x dx$

$$= \frac{[(3-1)\dots] \cdot [(4-1)(4-3)\dots]}{(3+4)(3+4-2)(3+4-4)(3+4-6)\dots}$$

$$= \frac{(3-1)(4-1)(4-3)}{7(7-2)(7-4)(7-6)} = \frac{2 \cdot \cancel{3} \cdot 1}{7 \cdot 5 \cdot \cancel{3} \cdot 1} = \frac{2}{35}$$



## \* Conversion Formulae —

As Reduction formulae are applied only on interval 0 to  $\frac{\pi}{2}$ .

Therefore for the intervals 0 to  $\pi$  and 0 to  $2\pi$  we have some conversion formulae.

$$\textcircled{1} \quad \int_0^{\pi} \sin^m(x) dx = \begin{cases} 2 \int_0^{\pi/2} \sin^m x dx & \text{where } m = \text{even} \\ 0 & \text{when } m = \text{odd} \end{cases}$$

$$\textcircled{2} \quad \int_0^{\pi} \cos^n(x) dx = \begin{cases} 2 \int_0^{\pi/2} \cos^n(x) dx & \text{when } n = \text{even} \\ 0 & \text{when } n = \text{odd} \end{cases}$$

$$\textcircled{3} \quad \int_0^{\pi} \sin^m x \cos^n x dx = \begin{cases} 2 \int_0^{\pi/2} \sin^m x \cos^n x dx & \text{when } n = \text{even} \\ 0 & \text{when } n = \text{odd} \end{cases}$$

It depends only on power of cos. function.

$$\text{Ex) } \int_0^{\pi} \sin^6 x dx$$

$$\Rightarrow \int_0^{\pi} \sin^6 x dx = 2 \int_0^{\pi/2} \sin^6 x dx \quad \text{since } n=6 \text{ is even.}$$

$$= 2 \left[ \frac{5 \cdot 3 \cdot 1 \cdot \pi}{6 \cdot 4 \cdot 2 \cdot 2} \right]$$

$$= \boxed{\frac{5\pi}{16}}$$

$$\text{Ex) } \int_0^{\pi} \cos^5 x dx = 0 \quad \text{since } n=5 \text{ is odd.}$$

\* Conversion formulae for 0 to  $2\pi$

$$\textcircled{1} \quad \int_0^{2\pi} \sin^m x dx = 4 \int_0^{\pi/2} \sin^m x dx \quad m = \text{even}$$

$$= 0 \quad m = \text{odd.}$$

$$\textcircled{2} \quad \int_0^{2\pi} \cos^n x dx = 4 \int_0^{\pi/2} \cos^n x dx \quad n = \text{even}$$

$$= 0 \quad n = \text{odd.}$$

$$\textcircled{3} \quad \int_0^{2\pi} \sin^m x \cos^n x dx = 4 \int_0^{\pi/2} \sin^m x \cos^n x dx$$

both  $m$  &  $n$  even

$$= 0 \quad \text{otherwise.}$$

$$\text{Ex) } \int_0^{2\pi} \sin^3 x \cos^4 x dx$$

$$= 0 \quad \begin{array}{l} \text{since } m=3 \text{ odd} \\ n=4 \text{ even.} \end{array}$$

$$\text{Ex) } \int_0^{2\pi} \sin^4 x \cos^6 x dx$$

$$= 4 \int_0^{\pi/2} \sin^4 x \cos^6 x dx \quad m=n \Rightarrow \text{even}$$

$$= 4 \left[ \frac{(4-1)(4-3) \cdots (6-1)(6-3)(6-5)}{(4+6)(10-2)(10-4)(10-6)(10-8) \cdots} \right]$$

$$= 4 \left[ \frac{3 \cdot 1 \cdot \cancel{8} \cdot \cancel{8} \cdot 1}{\cancel{2} \cdot 10 \cdot 8 \cdot \cancel{6} \cdot 4 \cdot 2} \right]$$

$$= 4 \left[ \frac{\cancel{48} \cdot 3}{256} \right] = \boxed{\frac{3}{64}}$$



Ex)  $\int_0^{\pi} x \sin^5 x \cos^2 x dx$ .

$\Rightarrow$  let  $I = \int_0^{\pi} x \sin^5 x \cos^2 x dx$ .

As  $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ .

$\therefore I = \int_0^{\pi} (\pi-x) \sin^5(\pi-x) \cos^2(\pi-x) dx$

but  $\sin(\pi-x) = \sin x$  and  $\cos(\pi-x) = -\cos x$

$\therefore I = \int_0^{\pi} (\pi-x) \sin^5 x (-\cos^2 x) dx$

$\therefore I = \int_0^{\pi} (\pi-x) \sin^5 x \cos^2 x dx \quad \because (-1)^2 = 1$

$\therefore I = \pi \int_0^{\pi} \sin^5 x \cos^2 x dx - \int_0^{\pi} x \sin^5 x \cos^2 x dx$

$I = 2\pi \int_0^{\pi/2} \sin^5 x \cos^2 x dx - I$

$\because \int_0^{\pi} \sin^m x \cos^n x dx = 2 \int_0^{\pi/2} \sin^m x \cos^n x dx$

$I + I = 2\pi \left[ \frac{(5-1)(5-3) \dots (2-1) \dots}{7(7-2)(7-4)(7-6) \dots} \right]$

$\cancel{2}I = \cancel{2}\pi \left[ \frac{4 \times 2 \times 1}{7 \times 5 \times 3 \times 1} \right]$

$I = \frac{8\pi}{105}$

Que) IF  $I_n = \int_{\pi/4}^{\pi/2} \cot^n \theta d\theta$  then prove that,

$I_n = \frac{1}{n-1} - I_{n-2}$ . Hence evaluate  $I_3$ .

$\Rightarrow$  let  $I_n = \int_{\pi/4}^{\pi/2} \cot^n \theta d\theta$

$$\therefore I_n = \int_{\pi/4}^{\pi/2} \cot^{(n-2+2)} \theta \, d\theta$$

$$I_n = \int_{\pi/4}^{\pi/2} \cot^{n-2} \theta \cdot \cot^2 \theta \, d\theta$$

But  $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta \Rightarrow \cot^2 \theta = \operatorname{cosec}^2 \theta - 1$

$$\therefore I_n = \int_{\pi/4}^{\pi/2} \cot^{n-2} \theta (\operatorname{cosec}^2 \theta - 1) \, d\theta$$

$$I_n = \int_{\pi/4}^{\pi/2} \cot^{n-2} \theta \cdot \operatorname{cosec}^2 \theta \, d\theta - \int_{\pi/4}^{\pi/2} \cot^{n-2} \theta \, d\theta$$

but  $I_n = \int_{\pi/4}^{\pi/2} \cot^n \theta \, d\theta$

$$\Rightarrow I_{n-2} = \int_{\pi/4}^{\pi/2} \cot^{n-2} \theta \, d\theta$$

$$\therefore I_n = \int_{\pi/4}^{\pi/2} \cot^{n-2} \theta \operatorname{cosec}^2 \theta \, d\theta - I_{n-2}$$

put  $\cot \theta = t$

$$\therefore \operatorname{cosec}^2 \theta \, d\theta = dt$$

When  $\theta = \pi/4 \Rightarrow t = \cot(\pi/4) = 1$

$\theta = \pi/2 \Rightarrow t = \cot(\pi/2) = 0$

$\therefore$  Above integral becomes

$$I_n = - \int_1^0 t^{n-2} \, dt - I_{n-2}$$

$$I_n = - \left[ \frac{t^{n-2+1}}{n-2+1} \right]_1^0 - I_{n-2}$$

$$I_n = - \left[ \frac{t^{n-1}}{n-1} \right]_1^0 - I_{n-2}$$



$$I_n = - \left[ \frac{0^{n-1}}{n-1} - \frac{1^{n-1}}{(n-1)} \right] - I_{n-2}$$

$$I_n = - \left( 0 - \frac{1}{n-1} \right) - I_{n-2}$$

$$\boxed{I_n = \frac{1}{n-1} - I_{n-2}} \longrightarrow (*)$$

Now to find  $I_3$ , put  $n=3$  in eqn (\*)

$$\therefore I_3 = \frac{1}{3-1} - I_{3-2}$$

$$I_3 = \frac{1}{2} - I_1 \longrightarrow (1)$$

Now find  $I_1$  for given interval,

$$\text{As } I_n = \int_{\pi/4}^{\pi/2} \cot^n \theta d\theta$$

$$n=1 \Rightarrow I_1 = \int_{\pi/4}^{\pi/2} \cot \theta d\theta$$

$$= \left[ \log(\sin \theta) \right]_{\pi/4}^{\pi/2}$$

$$= \log(\sin(\pi/2)) - \log \sin(\pi/4)$$

$$= \log(1) - \log\left(\frac{1}{\sqrt{2}}\right)$$

$$I_1 = 0 - \log\left(\frac{1}{\sqrt{2}}\right)$$

$$\boxed{I_1 = -\log\left(\frac{1}{\sqrt{2}}\right)}$$

put this value in eqn (1)

$$\therefore I_3 = \frac{1}{2} - \left( -\log\left(\frac{1}{\sqrt{2}}\right) \right)$$

$$\boxed{I_3 = \frac{1}{2} + \log\left(\frac{1}{\sqrt{2}}\right)}$$

required answer.

Que) If  $I_n = \int_0^{\pi/2} x^n \cos x dx$  then prove that

$$I_n = \left(\frac{\pi}{2}\right)^n - n(n-1) I_{n-2}.$$

$$\Rightarrow \text{Let } I_n = \int_0^{\pi/2} x^n \cos x dx.$$

By Parts Rule

$$\int_a^b u v dx = \left[ u \int v dx \right]_a^b - \int_a^b \left[ \frac{du}{dx} \int v dx \right] dx.$$

$$I_n = \left[ x^n \int \cos x dx \right]_0^{\pi/2} - \int_0^{\pi/2} \left( \frac{d}{dx} (x^n) \int \cos x dx \right) dx.$$

$$I_n = \left[ x^n \cdot \sin x \right]_0^{\pi/2} - \int_0^{\pi/2} (n x^{n-1} \sin x) dx.$$

$$I_n = \left[ \left(\frac{\pi}{2}\right)^n \cdot \sin(\pi/2) - 0^n \sin(0) \right] - n \int_0^{\pi/2} x^{n-1} \sin x dx$$

$$I_n = \left( \left(\frac{\pi}{2}\right)^n (1) - 0 \right) - n \int_0^{\pi/2} x^{n-1} \sin x dx$$

$$\therefore I_n = \left(\frac{\pi}{2}\right)^n - n \left[ \left[ x^{n-1} \int \sin x dx \right]_0^{\pi/2} - \int_0^{\pi/2} \left( \frac{d}{dx} (x^{n-1}) \int \sin x dx \right) dx \right]$$

$$I_n = \left(\frac{\pi}{2}\right)^n - n \left\{ \left[ x^{n-1} (-\cos x) \right]_0^{\pi/2} - \int_0^{\pi/2} (n-1) x^{n-2} (-\cos x) dx \right\}$$

$$I_n = \left(\frac{\pi}{2}\right)^n - n \left\{ - \left( x^{n-1} \cos x \right)_0^{\pi/2} + (n-1) \int_0^{\pi/2} x^{n-2} \cos x dx \right\}$$

$$\text{but } I_{n-2} = \int_0^{\pi/2} x^{n-2} \cos x dx.$$

$\therefore$



$$I_n = \left(\frac{\pi}{2}\right)^n - n \left[ -\left(\frac{\pi}{2}\right)^{n-1} \cos\left(\frac{\pi}{2}\right) - 0^{n-1} \cos(0) \right] + (n-1) I_{n-2}$$

As  $\cos(\pi/2) = 0$ ,  $\cos(0) = 1$

$$I_n = \left(\frac{\pi}{2}\right)^n - n \left[ -\left(\frac{\pi}{2}\right)^{n-1} (0) + 0(1) + (n-1) I_{n-2} \right]$$

$$I_n = \left(\frac{\pi}{2}\right)^n - n(n-1) I_{n-2} \therefore \text{Hence proved.}$$

Que If  $I_n = \int_0^{\pi/2} x^n \sin x dx$  then prove that

H.W.  
(for)

$$I_n = \left(\frac{\pi}{2}\right)^{n-1} - n(n-1) I_{n-2}.$$

Que) If  $I_n = \int_0^{\pi/4} \tan^n \theta d\theta$  then show that

$$I_n = \frac{1}{n-1} - I_{n-2}.$$

$$\Rightarrow \text{Let } I_n = \int_0^{\pi/4} \tan^n \theta d\theta.$$

$$I_n = \int_0^{\pi/4} \tan^{(n-2)+2} \theta d\theta.$$

$$I_n = \int_0^{\pi/4} \tan^{(n-2)} \theta \cdot \tan^2 \theta d\theta$$

As  $1 + \tan^2 \theta = \sec^2 \theta \Rightarrow \tan^2 \theta = \sec^2 \theta - 1$

$$\therefore I_n = \int_0^{\pi/4} \tan^{(n-2)} \theta \cdot (\sec^2 \theta - 1) d\theta$$

$$\therefore I_n = \int_0^{\pi/4} \tan^{n-2} \theta \cdot \sec^2 \theta d\theta - \int_0^{\pi/4} \tan^{n-2} \theta d\theta$$

$$\therefore I_n = \int_0^{\pi/4} \tan^{n-2} \theta \cdot \sec^2 \theta d\theta - I_{n-2}$$

put  $\tan \theta = t$   
 $\sec^2 \theta d\theta = dt$

When  $\theta = 0 \Rightarrow t = \tan(0) = 0$   
 $\theta = \pi/4 \Rightarrow t = \tan(\pi/4) = 1$

$$\therefore I_n = \int_0^1 t^{n-2} dt - I_{n-2}$$

$$I_n = \left[ \frac{t^{n-2+1}}{n-2+1} \right]_0^1 - I_{n-2}$$

$$I_n = \left[ \frac{t^{n-1}}{n-1} \right]_0^1 - I_{n-2}$$

$$\therefore I_n = \left[ \frac{1^{(n-1)}}{(n-1)} - \frac{0^{(n-1)}}{(n-1)} \right] - I_{n-2}$$

$$I_n = \left( \frac{1}{(n-1)} - 0 \right) - I_{n-2}$$

$$\therefore \boxed{I_n = \frac{1}{n-1} - I_{n-2}} \quad \text{Hence proved.}$$

Que) If  $I_n = \int_0^{\pi/4} \sec^n \theta d\theta$  then prove that

H.W.  
(TOT)

$$I_n = \frac{(\sqrt{2})^{n-2}}{(n-1)} + \frac{(n-2)}{(n-1)} I_{n-2}.$$

$\Rightarrow$  Given,  $I_n = \int_0^{\pi/4} \sec^n \theta d\theta$

$$\therefore I_n = \int_0^{\pi/4} \sec^{n-2} \theta \cdot \sec^2 \theta d\theta.$$

$\uparrow$                        $\uparrow$   
 $u$                        $v$



By parts.

$$\int_a^b uv dx = \left[ u \int v dx \right]_a^b - \int_a^b \left( \frac{du}{dx} \int v dx \right) dx.$$

$$\therefore I_n = \left[ \sec^{n-2} \theta \cdot \int \sec^2 \theta d\theta \right]_0^{\pi/4} - \int_0^{\pi/4} \left( \frac{d}{d\theta} (\sec^{n-2} \theta) \cdot \int \sec^2 \theta d\theta \right) d\theta.$$

$$\therefore I_n = \left[ \sec^{n-2} \theta \tan \theta \right]_0^{\pi/4} - \int_0^{\pi/4} ((n-2) \sec^{n-3} \theta \cdot \sec \theta \tan \theta \cdot \tan \theta) d\theta.$$

$$\therefore \int \sec^2 x dx = \sec x \tan x$$

$$\frac{d}{dx} (\sec x) = \sec x \tan x.$$

$$\frac{d}{dx} (x^n) = n x^{n-1}.$$

$$I_n = \left[ \sec^{n-2}(\pi/4) \tan(\pi/4) - \sec^{n-2}(0) \tan(0) \right] -$$

$$(n-2) \int_0^{\pi/4} \sec^{n-2} \theta \cdot \tan^2 \theta d\theta.$$

$$\text{but } \tan^2 \theta = \sec^2 \theta - 1, \sec(\pi/4) = \sqrt{2}, \tan(\pi/4) = 1$$

$$I_n = \left[ (\sqrt{2})^{n-2} (1) - 0 \right] - (n-2) \int_0^{\pi/4} \sec^{n-2} \theta \cdot (\sec^2 \theta - 1) d\theta.$$

$$I_n = (\sqrt{2})^{n-2} - (n-2) \int_0^{\pi/4} \sec^{n-2} \theta \cdot \sec^2 \theta d\theta + (n-2) \int_0^{\pi/4} \sec^{n-2} \theta d\theta.$$

$$I_n = (\sqrt{2})^{n-2} - (n-2) \int_0^{\pi/4} \sec^n \theta d\theta + (n-2) \int_0^{\pi/4} \sec^{n-2} \theta d\theta$$

$$\therefore I_n = (\sqrt{2})^{n-2} - (n-2) I_n + (n-2) I_{n-2}.$$

$$I_n + (n-2) I_n = (\sqrt{2})^{n-2} + (n-2) I_{n-2}.$$

$$(1+n-2) I_n = (\sqrt{2})^{n-2} + (n-2) I_{n-2}$$

$$(n-1) I_n = (\sqrt{2})^{n-2} + (n-2) I_{n-2}$$

$$\therefore I_n = \frac{(\sqrt{2})^{n-2} + (n-2) I_{n-2}}{(n-1)}$$

$$\boxed{I_n = \frac{(\sqrt{2})^{n-2}}{(n-1)} + \frac{(n-2) I_{n-2}}{(n-1)}}$$

Hence proved.



\* **Gamma Functions**:- The definite integral  $\int_0^{\infty} e^{-x} x^n dx$  is called as gamma function and it is given by

$$\int_0^{\infty} e^{-x} x^n dx = \Gamma(n+1)$$

\* The gamma function  $\Gamma n$  is read as gamma of n.

\* **Properties of Gamma functions**

①  $\Gamma(n+1) = n\Gamma n$  if n is positive integer, real no.

②  $\Gamma n = (n-1)!$  if n is a positive real number, integer.

③  $\Gamma p \cdot \Gamma(1-p) = \frac{\pi}{\sin p\pi}$   $0 < p < 1$

④  $\Gamma \frac{1}{2} = \sqrt{\pi}$

⑤  $\Gamma 0 = \infty$

⑥  $\Gamma 1 = 1$

⑦  $\Gamma n = 2 \int_0^{\infty} e^{-x^2} x^{2n-1} dx$

Ques) Evaluate  $\int_0^{\infty} \sqrt{x} e^{-x^3} dx$

$\Rightarrow$  Given,  $I = \int_0^{\infty} \sqrt{x} e^{-x^3} dx$ .

$\therefore I = \int_0^{\infty} e^{-x^3} x^{1/2} dx$

$\therefore$  substitute  $x^3 = t \Rightarrow x = t^{1/3}$

~~Differentiating w.r.t.  $x$~~   
 ~~$3x^2 dx = dt$~~

~~$dx = \frac{1}{3x^2} dt$~~

~~$dx =$~~

~~As  $x = t^{1/3}$~~

~~differentiating w.r.t.  $x$~~

~~$\left(\frac{dx}{dt}\right) = \frac{d}{dt} (t^{1/3})$~~

~~$1 = \frac{1}{3} t^{(1/3-1)} \frac{dt}{dx}$~~

~~$\therefore dx = \frac{1}{3} t^{-2/3} dt$~~

$x = t^{1/3}$

differentiating

$dx = \frac{1}{3} t^{(1/3-1)} dt$

$dx = \frac{1}{3} t^{-2/3} dt$

limits

When  $x = 0 \Rightarrow t = 0^3 = 0$

$x = \infty \Rightarrow t = \infty^3 = \infty$

convert the integral in terms of  $t$ .

$I = \int_0^{\infty} e^{-x^3} x^{1/2} dx$

$I = \int_0^{\infty} e^{-t} (t^{1/3})^{1/2} \cdot \left(\frac{1}{3} t^{-2/3} dt\right)$

$I = \frac{1}{3} \int_0^{\infty} e^{-t} t^{1/6} t^{-2/3} dt$



$$I = \frac{1}{3} \int_0^{\infty} e^{-t} t^{\frac{1}{3}-\frac{2}{3}} dt$$

$$I = \frac{1}{3} \int_0^{\infty} e^{-t} t^{-\frac{1}{3}} dt$$

Now by defn of gamma function,  $n = -\frac{1}{3}$

$$I = \frac{1}{3} \Gamma\left(-\frac{1}{3} + 1\right)$$

$$I = \frac{1}{3} \Gamma\left(\frac{2}{3}\right)$$

$$\boxed{I = \frac{1}{3} \sqrt{\pi}}$$

Que) Evaluate  $\int_0^{\infty} \sqrt[4]{x} \cdot e^{-\sqrt{x}} dx$ .

$$\Rightarrow \text{Let } I = \int_0^{\infty} \sqrt[4]{x} \cdot e^{-\sqrt{x}} dx$$

$$\therefore I = \int_0^{\infty} x^{1/4} \cdot e^{-x^{1/2}} dx$$

$$I = \int_0^{\infty} e^{-x^{1/2}} \cdot x^{1/4} dx$$

$$\text{Put } x^{1/2} = t \Rightarrow x = t^2$$

~~Differentiate w.r.t x~~

Differentiating it

$$dx = 2t dt$$

$$\text{When } x=0 \Rightarrow t = 0^{1/2} = 0$$

$$x=\infty \Rightarrow t = \infty^{1/2} = \infty$$

∴

$$\therefore I = \int_0^{\infty} e^{-x^{1/2}} x^{1/4} dx$$

$$\therefore I = \int_0^{\infty} e^{-t} (t^2)^{1/4} (2t dt)$$

$$\therefore I = 2 \int_0^{\infty} e^{-t} t^{1/2} t dt$$

$$I = 2 \int_0^{\infty} e^{-t} t^{(1/2+1)} dt \quad (\because a^m \cdot a^n = a^{m+n})$$

$$I = 2 \int_0^{\infty} e^{-t} t^{3/2} dt$$

By def<sup>n</sup> of gamma function

$$I = 2 \sqrt{\frac{3}{2} + 1}$$

$$\because \int_0^{\infty} e^{-x} x^n dx = \Gamma(n+1)$$

$$I = 2 \cdot \frac{3}{2} \sqrt{\frac{3}{2}}$$

$$\Gamma(n+1) = n \Gamma(n)$$

$$I = 2 \cdot \frac{3}{2} \sqrt{\frac{1}{2} + 1}$$

$$I = \cancel{2} \cdot \frac{3}{2} \cdot \cancel{\frac{1}{2}} \cdot \sqrt{\frac{1}{2}}$$

$$\boxed{I = \frac{3}{2} \sqrt{\pi}}$$

Que) Evaluate  $\int_0^{\infty} x^9 e^{-2x^2} dx$

$$\Rightarrow \text{Given, } I = \int_0^{\infty} e^{-2x^2} x^9 dx$$

$$\text{substitute } x^2 = t \Rightarrow \boxed{x = t^{1/2}}$$

As  $x = t^{1/2}$  on differentiating

$$dx = \frac{1}{2} t^{(1/2-1)} dt$$



$$\therefore dx = \frac{1}{2} t^{-1/2} dt$$

When  $x=0$ ,  $t=0^2=0$   
 $x=\infty$ ,  $t=\infty^2=\infty$ .

Now convert the integral in terms of  $t$ .

$$\therefore I = \int_0^{\infty} e^{-2x^2} x^9 dx$$

$$I = \int_0^{\infty} e^{-2t} (t^{1/2})^9 \left(\frac{1}{2} t^{-1/2}\right) dt$$

$$\therefore I = \frac{1}{2} \int_0^{\infty} e^{-2t} t^{9/2 - 1/2} dt$$

$$(a^m)^n = a^{mn}$$

$$I = \frac{1}{2} \int_0^{\infty} e^{-2t} t^{\left(\frac{9}{2} - \frac{1}{2}\right)} dt$$

$$a^m \cdot a^n = a^{m+n}$$

$$I = \frac{1}{2} \int_0^{\infty} e^{-2t} t^4 dt$$

By definition of gamma function  $\int_0^{\infty} e^{-kx} x^n dx = \frac{\Gamma(n+1)}{k^{n+1}}$

$$I = \frac{1}{2} \frac{\Gamma(4+1)}{2^{4+1}}$$

$$\therefore I = \frac{1}{2} \frac{\Gamma 5}{2^5} = \frac{4!}{2^6} = \frac{24}{64} = \frac{3}{8}$$

$$\Gamma n = (n-1)!$$

$n \in \mathbb{Z}^+$

$$\therefore \boxed{I = \frac{3}{8}}$$

Que)

H.W.

$\Rightarrow$

$$\int_0^{\infty} x^7 e^{-2x^2} dx$$

$$I = \frac{3}{16}$$

\* Example of the type  $\int \log\left(\frac{1}{x}\right) dx$  and  $\int \log(x) dx$

Que) Evaluate  $\int (x \log x)^4 dx$ .

$$\Rightarrow \text{Let } I = \int x^4 (\log x)^4 dx \quad (ab)^m = a^m b^m$$

Substitute  $\log x = -t$

$$(\log x)^4 = 4(\log x) : \quad e^{\log x} = e^{-t}$$

$$\boxed{x = e^{-t}}$$

on differentiating

$$dx = -e^{-t} dt$$

$$\text{When } x=0 \Rightarrow t = -\log(0) = -(-\infty) = \infty$$

$$x=1 \Rightarrow t = -\log(1) = 0$$

Now convert the integral in terms of  $t$ .

$$\therefore I = \int x^4 (\log x)^4 dx$$

$$\therefore I = \int_0^{\infty} (e^{-t})^4 (-t)^4 (-e^{-t} dt)$$

$$I = - \int_0^{\infty} e^{-4t} \cdot t^4 \cdot e^{-t} dt$$

$$\text{But } \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\therefore I = - \left[ - \int_0^{\infty} e^{-4t-t} \cdot t^4 dt \right]$$



$$\therefore I = + \int_0^{\infty} e^{-5t} \cdot t^4 dt.$$

By definition of gamma functions,

$$I = \frac{\Gamma(4+1)}{5^{4+1}}$$

$$I = \frac{\Gamma(5)}{5^5} = \frac{4!}{3125} = \frac{24}{3125}$$

Hence  $I = \frac{24}{3125}$

Que) Evaluate  $\int_0^1 (x \log x)^3 dx$

$$\Rightarrow I = -3/128.$$

\* Note:- For  $\log(1/x)$ , use the substitution  $\log(1/x) = t$ .

\* Examples of the type:  $-\int_0^{\infty} \frac{dx}{a^x}$

Que) Evaluate  $\int_0^{\infty} \frac{dx}{(3^4 x^2)}$

Que) Evaluate  $\int_0^{\infty} \frac{x^3}{3^x} dx$

$$\Rightarrow \text{Let } I = \int_0^{\infty} \frac{x^3}{3^x} dx$$

Substitute  $3 = e^m$

$$\therefore I = \int_0^{\infty} \frac{x^3}{(e^m)^x} dx$$

$$\therefore I = \int_0^{\infty} \frac{x^3}{e^{mx}} dx$$

$$I = \int_0^{\infty} e^{-mx} x^3 dx$$

By definition of gamma function

$$I = \frac{\Gamma(3+1)}{m^{3+1}}$$

$$\therefore \int_0^{\infty} e^{-kx} x^n dx = \frac{\Gamma(n+1)}{k^{n+1}}$$

$$I = \frac{\Gamma(4)}{m^4} \longrightarrow \textcircled{B}$$

but  $3 = e^m$

taking log on both sides

$$\log 3 = \log e^m$$

$$\log 3 = m \cdot \log(e)$$

$$\therefore \log(e) = 1$$

$$\log 3 = m(1)$$

$$\boxed{m = \log 3}$$

$\therefore e^{a^x} \textcircled{A}$  becomes

$$I = \frac{\Gamma(4)}{(\log 3)^4} = \frac{3!}{(\log 3)^4} = \frac{6}{(\log 3)^4}$$

$$\therefore \boxed{I = \frac{6}{(\log 3)^4}}$$

Que) Evaluate  $\int_0^{\infty} \frac{x^4}{4^x} dx \Rightarrow I = \frac{24}{(\log 4)^5}$   
H.W.

Que) Evaluate  $\int_0^{\infty} \frac{dx}{3^{4x^2}}$



let  $I = \int_0^{\infty} \frac{dx}{a^4 x^2}$

put  $a = e^m$

$\therefore I = \int_0^{\infty} \frac{dx}{(e^m)^4 x^2}$

$\therefore I = \int_0^{\infty} \frac{dx}{e^{4mx^2}}$

$I = \int_0^{\infty} e^{-4mx^2} dx \longrightarrow \textcircled{1}$

Substitute  $x^2 = t$

$\Rightarrow [x = t^{1/2}]$

on differentiating

$dx = \frac{1}{2} t^{(1/2-1)} dt$

$dx = \frac{1}{2} t^{-1/2} dt$

Above integral  $\textcircled{1}$  becomes

When  $x = 0 \Rightarrow t = 0^2 = 0$

$x = \infty \Rightarrow t = \infty^2 = \infty$

$\therefore I = \int_0^{\infty} e^{-4mt} \left( \frac{1}{2} t^{-1/2} dt \right)$

$I = \frac{1}{2} \int_0^{\infty} e^{-4mt} t^{-1/2} dt$

By def<sup>n</sup> of gamma function,

$I = \frac{1}{2} \frac{\Gamma(-\frac{1}{2}+1)}{(4m)^{(\frac{1}{2}+1)}}$

$\therefore \int_0^{\infty} e^{-kt} t^n dt = \frac{\Gamma(n+1)}{k^{n+1}}$

$$\therefore I = \frac{1}{2} \cdot \frac{\sqrt{1/2}}{(4m)^{1/2}}$$

$$\text{But } \sqrt{1/2} = \sqrt{\pi}$$

$$\therefore I = \frac{1}{2} \frac{\sqrt{\pi}}{\sqrt{4m}} = \frac{1}{2} \cdot \frac{\sqrt{\pi}}{2\sqrt{m}} = \frac{1}{4} \frac{\sqrt{\pi}}{\sqrt{m}}$$

$$\text{but } 3 = e^m$$

$$\therefore \log 3 = \log e^m$$

$$\log 3 = m \cdot \log e$$

$$(\because \log e = 1)$$

$$\boxed{m = \log 3}$$

$$\therefore I = \frac{1}{4} \frac{\sqrt{\pi}}{\sqrt{\log 3}}$$

Hence

$$\boxed{I = \frac{\sqrt{\pi}}{4\sqrt{\log 3}}}$$



\* **Beta Functions:**— Many complex integrals can be reduced to expressions involving the beta function.

The definite integral  $\int_0^1 x^m (1-x)^n dx$   $m > 0, n > 0$  is a function of  $m$  and  $n$  is called as Beta function. It is denoted by  $\beta(m, n)$

$$\therefore \int_0^1 x^m (1-x)^n dx = \beta(m+1, n+1).$$

\* Note:— The Beta functions of negative numbers are not defined.

\* **Properties of Beta Functions:**

$$(1) \quad \beta(m, n) = \beta(n, m)$$

$$(2) \quad \int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

$$(3) \quad \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx = \beta(m, n)$$

(4) **Rel<sup>n</sup> bet<sup>n</sup> beta and Gamma function**

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$(5) \quad \Gamma(p) \cdot \Gamma(1-p) = \frac{\pi}{\sin(p\pi)} \quad 0 < p < 1$$

(6) **Duplication formula of Gamma Functions:**

$$\Gamma(m) \cdot \Gamma\left(m + \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{m-1}} \cdot \Gamma(2m).$$

Que) Evaluate  $\int_0^1 x^3 (1-\sqrt{x})^5 dx$ .

$$\Rightarrow \text{Let } I = \int_0^1 x^3 (1-\sqrt{x})^5 dx$$

$$\therefore I = \int_0^1 x^3 (1-x^{\frac{1}{2}})^5 dx$$

Step ① By substitution  $x^{\frac{1}{2}} = t$

Step ②

$$\Rightarrow x = t^2$$

Step ③

on Differentiation

$$dx = 2t dt$$

\* Step ④ Limits:

$$\text{When } x=0 \Rightarrow t=0^{\frac{1}{2}}=0$$

$$x=1 \Rightarrow t=1^{\frac{1}{2}}=1$$

\* Step ⑤ Convert the integrals in terms of  $t$

$$\therefore I = \int_0^1 (t^2)^3 (1-t)^5 2t dt$$

$$I = 2 \int_0^1 t^6 (1-t)^5 \cdot t dt$$

$$I = 2 \int_0^1 t^7 (1-t)^5 dt$$

Here  $m=7, n=5$

By definition of Beta Function

$$I = 2 \beta(7+1, 5+1)$$

$$I = 2 \beta(8, 6)$$

$$I = 2 \cdot \frac{1}{8} \cdot \frac{1}{6}$$

$$\frac{1}{8+6}$$

But  $T_n = (n-1)!$  when  $n \in \mathbb{Z}^+$

$$\int_0^1 x^m (1-x)^n dx = \beta(m+1, n+1)$$



$$\therefore F = 2 \cdot \frac{7! \cdot 6!}{14!}$$

$$I = 2 \cdot \frac{7! \cdot 5!}{13!} = \frac{2 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7!} = \frac{1}{5148}$$

$$\boxed{I = \frac{1}{5148}}$$

Que) Evaluate  $\int_0^1 \frac{dx}{\sqrt[3]{1-x^3}} \Rightarrow I = \frac{2\pi}{3\sqrt{3}}$

H.W.  
(TUT)

$\Rightarrow$  Hint.  $I = \int_0^1 (1-x^3)^{-1/3} dx$  put  $x^3 = t$ .

Que) Prove that,  $\int_0^1 x^{m-1} (1-x^2)^{n-1} dx = \frac{1}{2} \beta\left(\frac{m}{2}, n\right)$

$\Rightarrow$  let  $I = \int_0^1 x^{m-1} (1-x^2)^{n-1} dx$

Substitute  $x^2 = t$   
i.e.  $x = t^{1/2}$

on differentiating

$$dx = \frac{1}{2} t^{-1/2} dt$$

limits:

when  $x=0 \Rightarrow t = 0^2 = 0$

$x=1 \Rightarrow t = 1^2 = 1$

$$\therefore I = \int_0^1 x^{m-1} (1-x^2)^{n-1} dx$$

$$I = \int_0^1 (t^{1/2})^{m-1} (1-t)^{n-1} \frac{1}{2} t^{-1/2} dt$$

$$\therefore I = \frac{1}{2} \int_0^1 t^{\frac{(m-1)}{2}} t^{-1/2} (1-t)^{n-1} dt$$



$$I = \frac{1}{2} \int_0^1 t^{\left(\frac{m-1}{2}-\frac{1}{2}\right)} (1-t)^{n-1} dt$$

$$I = \frac{1}{2} \int_0^1 t^{\frac{m-2}{2}} (1-t)^{n-1} dt$$

By def<sup>n</sup> of Beta functions,

$$I = \frac{1}{2} \beta\left(\frac{m-2}{2}+1, n-1+1\right)$$

$$\therefore I = \frac{1}{2} \beta\left(\frac{m-2+2}{2}, n\right)$$

$$I = \frac{1}{2} \beta\left(\frac{m}{2}, n\right)$$

$$I = \text{R.H.S.}$$

Hence  $\int_0^1 x^{m-1} (1-x^2)^{n-1} dx = \frac{1}{2} \beta\left(\frac{m}{2}, n\right).$

Que) Evaluate  $\int_0^1 \frac{x^2 (1-x^2)^{3/2}}{3} dx$

$$\Rightarrow I = \frac{\pi}{96}$$

Que) Evaluate  $\int_0^{2a} x^{7/2} (2a-x)^{-1/2} dx$

$\Rightarrow$  Substitute  $x = 2at \Rightarrow t = \frac{x}{2a}$   
on differentiating  
 $dx = 2a dt$

Limits:

when  $x=0 \Rightarrow t = \frac{0}{2a} = 0$

when  $x = 2a$ ,  $\therefore t = \frac{2a}{2a} = 1$

$\therefore$  The integral becomes,

$$I = \int_0^{2a} x^{7/2} (2a-x)^{-1/2} dx$$

$$I = \int_0^1 (2at)^{7/2} (2a-2at)^{-1/2} 2a dt$$

$$\therefore I = \int_0^1 (2a)^{7/2} t^{7/2} \cdot (2a)^{-1/2} (1-t)^{-1/2} \cdot 2a dt$$

$$I = 2a \cdot (2a)^{7/2} (2a)^{-1/2} \int_0^1 t^{7/2} (1-t)^{-1/2} dt$$

$$I = 2a^{(1+7/2-1/2)} \int_0^1 t^{7/2} (1-t)^{-1/2} dt$$

$$I = (2a)^4 \int_0^1 t^{7/2} (1-t)^{-1/2} dt$$

$$m = 7/2, n = -1/2$$

$\therefore$  By def<sup>n</sup> of ~~gamma~~ <sup>Beta</sup> function,  $\int_0^1 x^m (1-x)^n dx = \beta(m+1, n+1)$

$$I = (2a)^4 \beta\left(\frac{7}{2}+1, \frac{-1}{2}+1\right) = \beta(m+1, n+1)$$

$$I = (2a)^4 \beta\left(\frac{9}{2}, \frac{1}{2}\right)$$

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$I = (2a)^4 \frac{\Gamma(9/2) \cdot \Gamma(1/2)}{\Gamma(9/2+1/2)}$$

$$I = \frac{2^4 \cdot a^4 \cdot \left(\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}\right) (\sqrt{\pi})}{\sqrt{5}}$$

But  $\Gamma n = (n-1)!$   
 $n \in \mathbb{Z}^+$

$$I = 16 \cdot 4 \cdot \frac{105\pi}{16} \cdot \frac{1}{4!}$$

$$I = \frac{105\pi \cdot 4}{24}$$

Que) Evaluate  $\int_2^5 \sqrt{(x-2)^7 (5-x)^9} dx$

$$\Rightarrow \text{let } I = \int_2^5 \sqrt{(x-2)^7 (5-x)^9} dx$$

$$I = \int_2^5 (x-2)^{7/2} (5-x)^{9/2} dx \rightarrow \textcircled{1}$$

substitute  $x = (\text{upper limit} - \text{lower limit})t + \text{lower limit}$

$$\therefore x = (5-2)t + 2$$

$$\boxed{x = 3t + 2} \implies 3t = x - 2$$

on Differentiating

$$dx = 3dt$$

$$\boxed{t = \frac{x-2}{3}}$$

limits

$$\text{When } x=2 \Rightarrow t = \frac{2-2}{3} = \frac{0}{3} = 0$$

$$\text{When } x=5 \Rightarrow t = \frac{5-2}{3} = \frac{3}{3} = 1$$

$\therefore$  from  $\textcircled{1}$

$$I = \int_2^5 (x-2)^{7/2} (5-x)^{9/2} dx$$

$$I = \int_0^1 (3t+2-2)^{7/2} (5-(3t+2))^{9/2} 3dt$$



$$\therefore I = \int_0^1 (3t)^{7/2} (5-3t-2)^{9/2} \cdot 3 dt$$

$$I = \int_0^1 3^{7/2} t^{7/2} (3-3t)^{9/2} \cdot 3 dt$$

$$I = 3^{7/2} \cdot 3 \int_0^1 t^{7/2} \cdot 3^{9/2} (1-t)^{9/2} dt$$

$$I = 3^{7/2} \cdot 3 \cdot 3^{9/2} \int_0^1 t^{7/2} (1-t)^{9/2} dt$$

$$\frac{7}{2} + 1 + \frac{9}{2} = 9$$

$$m = 7/2 \text{ and } n = 9/2$$

$$\therefore I = 3^9 \cdot B\left(\frac{7}{2}+1, \frac{9}{2}+1\right)$$

$$\int_0^1 x^m (1-x)^n dx = B(m+1, n+1)$$

$$I = 3^9 \cdot B\left(\frac{9}{2}, \frac{11}{2}\right)$$

$$B(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}$$

$$I = 19683 \cdot \frac{\Gamma(9/2) \Gamma(11/2)}{\Gamma(9/2 + 11/2)}$$

$$I = 19683 \cdot \frac{\left(\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}\right) \cdot \left(\frac{9}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}\right)}{\sqrt{10}}$$

$$I = 19683 \cdot \frac{\left(\frac{105\sqrt{\pi}}{16}\right)}{9!} \cdot \frac{\left(\frac{945\sqrt{\pi}}{32}\right)}{9!}$$

$$I = \frac{19683 \times 99225 \pi}{512 \times 362880} = 10.511 \pi$$

$$\boxed{I = 10.511 \pi}$$

Poschier Ex) ①  $\int_3^7 (x-3)^{1/4} (7-x)^{1/4} dx \Rightarrow I = 8 B\left(\frac{5}{4}, \frac{5}{4}\right)$

②  $\int_3^5 (x-3)^{1/2} (5-x)^{1/2} dx \Rightarrow I = \frac{\pi}{2}$

- \* **D.U.I.S:-** Differentiation Under Integral sign or briefly D.U.I.S is an effective technique used in evaluation of real definite integrals. When a definite Integral

$$I = \int_a^b f(x, \alpha) dx,$$

which is to be integrated w.r.t. variable  $x$  and contains other parameter  $\alpha$ . then

we differentiate the given integral w.r.t. ' $\alpha$ ' by using D.U.I.S.

- \* **Rule ①:-** Integral with constant limits:

$$\text{If } I(\alpha) = \int_a^b f(x, \alpha) dx,$$

where  $a$  and  $b$  are constants then

$$\frac{dI}{d\alpha} = \int_a^b \frac{\partial}{\partial \alpha} (f(x, \alpha)) dx.$$

Hence  $\frac{d}{d\alpha}$  becomes  $\frac{\partial}{\partial \alpha}$  (partial derivative) when

taken under integral sign.

Que) Prove that  $\int_0^1 \frac{x^a - 1}{\log x} dx = \log(1 + a)$ ,  $a \neq 0$ .

$$\Rightarrow \text{Let } I = \int_0^1 \frac{x^a - 1}{\log x} dx.$$

Here  $dx$  represents that variable of integration is ' $x$ '. But it is not possible to integrate w.r.t. ' $x$ '. Therefore we will consider another variable other than  $x$ .



Let's consider 'a'.

$$\therefore I(a) = \int_0^1 \frac{x^a - 1}{\log x} dx \quad \text{--- (1)}$$

Differentiate w.r.t. 'a'.

$$\frac{d}{da}(I) = \frac{d}{da} \int_0^1 \frac{x^a - 1}{\log x} dx$$

It can be written as 'I'.

By using rule of DUIS.

$$\textcircled{1} \frac{dI}{da} = \int_0^1 \frac{\partial}{\partial a} \left[ \frac{x^a - 1}{\log x} \right] dx$$

with respect to 'a' the variable  $x$  becomes constant.

$$\text{Also } \frac{d}{dx}(a^x) = a^x \log a$$

$$\Rightarrow \frac{d}{da}(x^a) = x^a \log x$$

Here derivative w.r.t.  $a$  and  $x$  is constant.

$$\therefore \frac{dI}{da} = \int_0^1 \frac{1}{\log x} \frac{\partial}{\partial a} (x^a - 1) dx$$

$$\frac{dI}{da} = \int_0^1 \frac{1}{\log x} [x^a \log x - 0] dx$$

$$\frac{dI}{da} = \int_0^1 x^a dx$$

Now the variable of integration is  $x$  and it is possible to integrate w.r.t.  $x$ .

$$\frac{dI}{da} = \left[ \frac{x^{a+1}}{a+1} \right]_0^1$$

$$\therefore \int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\therefore \frac{dI}{da} = \left[ \frac{1^{a+1}}{a+1} - \frac{0^{a+1}}{a+1} \right]$$

$$\therefore \frac{dI}{da} = \frac{1}{a+1} - \frac{0}{a+1} \quad \because 1^{a+1} = 1$$

$$\therefore \frac{dI}{da} = \frac{1}{a+1}$$

$$\therefore dI = \left( \frac{1}{a+1} \right) da$$

Variable separable form.

on integration

$$\int dI = \int \frac{da}{a+1} + C$$

$$\therefore I = \log(a+1) + C \longrightarrow \textcircled{2}$$

Now to find the value of constant 'c' we will consider the value of 'a' for which given integral (1) becomes zero.

$\therefore$  Given integral is

$$I(a) = \int_0^1 \frac{x^a - 1}{\log x} dx$$

put  $a=0$

$$I(0) = \int_0^1 \frac{x^0 - 1}{\log x} dx$$

$$I(0) = \int_0^1 \frac{1-1}{\log x} dx$$

$$\boxed{I(0) = 0}$$



from eqn ②

$$I(a) = \log(a+1) + C$$

put  $a=0$

$$I(0) = \log(0+1) + C$$

$$0 = \log(1) + C$$

$$C = 0 + C$$

$$\boxed{C=0}$$

from eqn ②

$$\boxed{I(a) = \log(a+1)}$$

Que) Prove that  $\int_0^1 \frac{x^a - x^b}{\log x} dx = \log\left(\frac{a+1}{b+1}\right)$  :  $a > 0, b > 0$   
(TUT)

Que) Prove that  $\int_0^\infty \frac{e^{-\alpha x} \sin x}{x} dx = \cot^{-1}(\alpha)$ .

$$\Rightarrow \text{let } I = \int_0^\infty \frac{e^{-\alpha x} \sin x}{x} dx \longrightarrow \textcircled{1}$$

Here  $dx$  represents that, variable of integration is  $x$ . but it is not possible to integrate w.r.t.  $x$ . Therefore we will consider the given integral of another variable other than  $x$ .

Let's consider the given integral is of variable  $\alpha$ .

$\therefore$  Differentiating w.r.t.  $\alpha$ .

$$\frac{dI}{d\alpha} = \frac{d}{d\alpha} \int_0^\infty \frac{e^{-\alpha x} \sin x}{x} dx$$

Now according to Difs Rule  $\frac{d}{d\alpha}$  becomes partial differentiation when it is taken under integral sign.



$$\frac{dI}{d\alpha} = \int_0^{\infty} \frac{\partial}{\partial \alpha} \left[ \frac{e^{-\alpha x} \sin x}{x} \right] dx$$

$$\therefore \frac{dI}{d\alpha} = \int_0^{\infty} \frac{\sin x}{x} \cdot \frac{\partial}{\partial \alpha} (e^{-\alpha x}) dx$$

$$\therefore \frac{dI}{d\alpha} = \int_0^{\infty} \frac{\sin x}{x} \cdot \frac{e^{-\alpha x}}{x} (-x) dx$$

$$\frac{d}{dx} (e^{\alpha x}) = \alpha \cdot e^{\alpha x}$$

$$\Rightarrow \frac{d}{d\alpha} (e^{\alpha x}) = x \cdot e^{\alpha x}$$

$$\therefore \frac{dI}{d\alpha} = - \int_0^{\infty} e^{-\alpha x} \sin x dx$$

By using,  $\int_0^{\infty} e^{-ax} \sin bx dx = \frac{b}{a^2 + b^2}$

Here  $a = \alpha$  and  $b = 1$

$$\therefore \frac{dI}{d\alpha} = - \left[ \frac{1}{\alpha^2 + 1} \right]$$

$$dI = - \frac{1}{\alpha^2 + 1} d\alpha \quad \dots \text{v.s. form}$$

on integration

$$\int dI = - \int \frac{d\alpha}{\alpha^2 + 1} + C$$

$$\boxed{I = -\tan^{-1}(\alpha) + C} \longrightarrow \textcircled{2}$$

Now to find the value of integration constant  $C$  we need to put the value of  $\alpha$  such that the integral ① becomes zero.

As  $I(x) = \int_0^{\infty} \frac{e^{-\alpha x} \sin x}{x} dx$

put  $\alpha = \infty$

$$I(\infty) = \int_0^{\infty} \frac{e^{-\infty(x)} \sin x}{x} dx$$

$$I(\infty) = 0$$

$$(e^{-\infty} = 0)$$

put this value of  $\alpha$  in eqn ②

$$\therefore I(\alpha) = -\tan^{-1}(\alpha) + C \quad \longrightarrow ③$$

put  $\alpha = \infty$

$$I(\infty) = -\tan^{-1}(\infty) + C$$

$$0 = -\frac{\pi}{2} + C$$

$$\boxed{C = \frac{\pi}{2}}$$

put this value of  $C$  in eqn ②

$$\therefore I = -\tan^{-1}(\alpha) + \frac{\pi}{2}$$

$$I = \frac{\pi}{2} - \tan^{-1}(\alpha)$$

$$\boxed{I = \cot^{-1}(\alpha)}$$

$\therefore$  Hence proved.



\* Rule-II Integral with Limits as Functions of the Parameter

IF  $I(\alpha) = \int_{a(\alpha)}^{b(\alpha)} f(x, \alpha) dx$ , where  $a$  and  $b$  both are functions of parameter  $\alpha$ , then,

$$\frac{dI}{d\alpha} = \int_{a(\alpha)}^{b(\alpha)} \frac{\partial}{\partial \alpha} f(x, \alpha) dx + f(b, \alpha) \frac{db}{d\alpha} - f(a, \alpha) \frac{da}{d\alpha}$$

Que) Show that  $\phi(a) = \int_{\pi/6a}^{\pi/2a} \frac{\sin(ax)}{x} dx$  is independent of  $a$ .

⇒ Given,

$$\phi(a) = \int_{\pi/6a}^{\pi/2a} \frac{\sin(ax)}{x} dx$$

Differentiating w.r.t. 'a'

Here we are differentiating w.r.t. 'a' and the limits are also in terms of 'a'. Therefore we use Rule II of DUIS.

$$\therefore \frac{d\phi}{da} = \int_{\pi/6a}^{\pi/2a} \frac{\partial}{\partial a} \left( \frac{\sin(ax)}{x} \right) dx + \frac{\sin\left(\cancel{a} \cdot \frac{\pi}{2\cancel{a}}\right)}{\left(\frac{\pi}{2\cancel{a}}\right)} \cdot \frac{d}{da} \left( \frac{\pi}{2a} \right) -$$

$$\frac{\sin\left(\cancel{a} \cdot \frac{\pi}{6\cancel{a}}\right)}{\left(\frac{\pi}{6\cancel{a}}\right)} \cdot \frac{d}{da} \left( \frac{\pi}{6a} \right)$$

$$\text{But } \frac{d}{da} \left( \frac{1}{a} \right) = -\frac{1}{a^2}, \quad \sin(\pi/2) = 1, \quad \sin(\pi/6) = \frac{1}{2}$$

$$\begin{aligned} \therefore \frac{d\phi}{da} &= \int_{\pi/6a}^{\pi/2a} \frac{1}{x} \cdot [\cos(ax) \cdot \cancel{x}] dx + \frac{\sin(\pi/2)}{\left(\frac{\pi}{2a}\right)} \cdot \frac{\pi}{a} \left( -\frac{1}{a^2} \right) \\ &\quad - \frac{\sin(\pi/6)}{\left(\frac{\pi}{6a}\right)} \cdot \frac{\pi}{a} \left( -\frac{1}{a^2} \right) \end{aligned}$$

$$\therefore \frac{d\phi}{da} = \int_{\pi/6a}^{\pi/2a} \cos(ax) dx - \frac{1}{a} \sin\left(\frac{\pi}{2}\right) + \frac{1}{a} \sin\left(\frac{\pi}{6}\right)$$

$$\therefore \frac{d\phi}{da} = \left[ \frac{\sin(ax)}{a} \right]_{\pi/6a}^{\pi/2a} - \frac{1}{a} \sin\left(\frac{\pi}{2}\right) + \frac{1}{a} \sin\left(\frac{\pi}{6}\right)$$

$$\frac{d\phi}{da} = \frac{\sin(\cancel{a} \cdot \frac{\pi}{2\cancel{a}})}{a} - \frac{\sin(\cancel{a} \cdot \frac{\pi}{6\cancel{a}})}{a} - \frac{1}{a} \sin(\pi/2) + \frac{1}{a} \sin(\pi/6)$$

$$\frac{d\phi}{da} = \frac{1 \cancel{\sin}(\pi/2)}{a} - \frac{1 \cancel{\sin}(\pi/6)}{a} - \frac{1 \cancel{\sin}(\pi/2)}{a} + \frac{1 \cancel{\sin}(\pi/6)}{a}$$

$$\frac{d\phi}{da} = 0$$

$$\Rightarrow \phi = \text{constant} \quad \therefore \frac{d}{dx}(\text{constant}) = 0$$

$$\Rightarrow \int a dx = \text{constant}$$

$\therefore \phi(a)$  is independent of  $a$ .

Que) IF  $f(x) = \int_a^x (x-t)^2 G(t) dt$  then show that

$$\frac{d^3 f}{dx^3} - 2G(x) = 0$$

$$\Rightarrow \text{Given } f(x) = \int_a^x (x-t)^2 G(t) dt$$

Differentiating w.r.t.  $x$

$$\frac{df}{dx} = \frac{d}{dx} \int_a^x (x-t)^2 G(t) dt$$

By Rule II of DUIS

$$\frac{df}{dx} = \int_a^x \frac{\partial}{\partial x} (x-t)^2 G(t) dt + (x-x)^2 G(x) \frac{d}{dx}(x) - (x-a)^2 G(a) \frac{d}{dx}(a)$$



$$\therefore \frac{df}{dx} = \int_a^x 2(x-t)(1-0)G(t)dt + 0 - 0$$

$$\therefore \frac{df}{dx}(a) = 0$$

$$\therefore \frac{df}{dx} = 2 \int_a^x (x-t)G(t)dt$$

Differentiate w.r.t.  $x$  again.

$$\frac{d^2f}{dx^2} = 2 \frac{d}{dx} \int_a^x (x-t)G(t)dt$$

Again by using Rule II of DUIS.

$$\frac{d^2f}{dx^2} = 2 \left\{ \int_a^x \frac{\partial}{\partial x} (x-t)G(t)dt + (x-x)G(x) \frac{d}{dx}(x) - (x-a)G(a) \frac{d}{dx}(a) \right\}$$

$$\therefore \frac{d^2f}{dx^2} = 2 \left\{ \int_a^x (1-0)G(t)dt + 0 - 0 \right\}$$

$$\frac{d^2f}{dx^2} = 2 \int_a^x G(t)dt$$

Differentiating w.r.t.  $x$  again

$$\frac{d^3f}{dx^3} = 2 \frac{d}{dx} \int_a^x G(t)dt$$

$$\frac{d^3f}{dx^3} = 2 \left\{ \int_a^x \frac{\partial}{\partial x} G(t)dt + G(x) \frac{d}{dx}(x) - G(a) \frac{d}{dx}(a) \right\}$$

$$\frac{d^3f}{dx^3} = 2 \left\{ \int_a^x 0 dt + G(x)(1) - 0 \right\}$$

$$\frac{d^3f}{dx^3} = 2G(x)$$

|                                 |               |
|---------------------------------|---------------|
| $\frac{d^3f}{dx^3} - 2G(x) = 0$ | Hence proved. |
|---------------------------------|---------------|



## Error Functions

\* ~~Error~~ Function - The error function of  $x$  denoted by  $\text{erf}(x)$  and is defined as

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2} du.$$

\* ~~Complementary~~ <sup>erf</sup> Error Function: - The ~~complementary~~ error function of  $x$  is denoted by  $\text{erfc}(x)$  and is defined as

$$\text{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-u^2} du.$$

\* Alternate Error function:-

$$\text{erf}(x) = \frac{1}{\sqrt{\pi}} \int_0^{x^2} e^{-u} u^{-1/2} du.$$

\* Alternate Complementary Error function:

$$\text{erfc}(x) = \frac{1}{\sqrt{\pi}} \int_{x^2}^{\infty} e^{-u} u^{-1/2} du.$$

\* Note: Either ① or ③ can be used for  $\text{erf}(x)$  according to the need of problem.

\* Some Properties of Error Functions:-

①  $\text{erf}(\infty) = 1$

②  $\text{erf}(0) = 0$

③  $\text{erf}(x) + \text{erfc}(x) = 1$

④ Error Function is an odd function .

$$\operatorname{erf}(-x) = -\operatorname{erf}(x).$$

⑤ In series,  $\operatorname{erf}(x)$  can be written as,

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \left[ x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \dots \right]$$

This series is uniformly convergent and hence  $\operatorname{erf}(x)$  is continuous function of  $x$ .

Que) show that  $\int_0^b e^{-x^2} dx = \frac{\sqrt{\pi}}{2} [\operatorname{erf}(b) - \operatorname{erf}(a)]$   $(\because 0 < a < b < \infty)$ .

$\Rightarrow$  From definition of  $\operatorname{erf}(x)$ , we have.

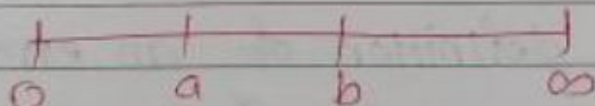
$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2} du.$$

put  $x = \infty$ .

$$\operatorname{erf}(\infty) = \frac{2}{\sqrt{\pi}} \int_0^{\infty} e^{-u^2} du$$

but  $0 < a < b < \infty$  and  $\operatorname{erf}(\infty) = 1$

$$\therefore 1 = \frac{2}{\sqrt{\pi}} \left[ \int_0^a e^{-u^2} du + \int_a^b e^{-u^2} du + \int_b^{\infty} e^{-u^2} du \right]$$



$$\therefore 1 = \frac{2}{\sqrt{\pi}} \int_0^a e^{-u^2} du + \frac{2}{\sqrt{\pi}} \int_a^b e^{-u^2} du + \frac{2}{\sqrt{\pi}} \int_b^{\infty} e^{-u^2} du.$$



$$1 = \operatorname{erf}(a) + \frac{2}{\sqrt{\pi}} \int_a^b e^{-u^2} du + \operatorname{erfc}(b)$$

$$\therefore \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2} du \quad \& \quad \operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-u^2} du$$

$$\therefore 1 - \operatorname{erfc}(b) = \operatorname{erf}(a) + \frac{2}{\sqrt{\pi}} \int_a^b e^{-u^2} du$$

$$\text{But } \operatorname{erf}(b) + \operatorname{erfc}(b) = 1$$

$$\operatorname{erfc}(b) = 1 - \operatorname{erf}(b)$$

$$\therefore 1 - (1 - \operatorname{erf}(b)) = \operatorname{erf}(a) + \frac{2}{\sqrt{\pi}} \int_a^b e^{-u^2} du$$

$$1 - 1 + \operatorname{erf}(b) = \operatorname{erf}(a) + \frac{2}{\sqrt{\pi}} \int_a^b e^{-u^2} du$$

$$\operatorname{erf}(b) - \operatorname{erf}(a) = \frac{2}{\sqrt{\pi}} \int_a^b e^{-u^2} du$$

$$\therefore \boxed{\int_a^b e^{-u^2} du = \frac{\sqrt{\pi}}{2} [\operatorname{erf}(b) - \operatorname{erf}(a)]}$$

Hence proved

Que) Prove that  $\frac{d}{dx} [\operatorname{erf}(x)] = \frac{2}{\sqrt{\pi}} e^{-x^2}$ .

HW.

$\Rightarrow$  From definition of an error function,

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2} du$$

Differentiating w.r.t.  $x$ .

$$\frac{d}{dx} [\operatorname{erf}(x)] = \frac{2}{\sqrt{\pi}} \frac{d}{dx} \int_0^x e^{-u^2} du$$

$\therefore$  We are differentiating w.r.t.  $x$  and upper limit of integration is in terms of  $x$ , Therefore we have to apply Rule-II of DIFS.

$$\frac{d}{dx} [\operatorname{erf}(x)] = \frac{2}{\sqrt{\pi}} \left\{ \int_0^x \frac{\partial}{\partial x} (e^{-u^2} du) + e^{-x^2} \frac{d}{dx} (x) - e^{-0^2} \frac{d}{dx} (0) \right\}$$

$$= \frac{2}{\sqrt{\pi}} \left\{ \int_0^x 0 du + e^{-x^2} (1) - 0 \right\}$$

$$\boxed{\frac{d}{dx} [\operatorname{erf}(x)] = \frac{2}{\sqrt{\pi}} e^{-x^2}} \quad \therefore \text{Hence proved.}$$

Que) Show that  $\int_0^{\infty} e^{-x^2-2bx} dx = \frac{\sqrt{\pi}}{2} e^{b^2} [1 - \operatorname{erf}(b)]$ .

$\Rightarrow$  Consider  
L.H.S. =  $\int_0^{\infty} e^{-x^2-2bx} dx$

$$\text{L.H.S.} = \int_0^{\infty} e^{-x^2-2bx-b^2+b^2} dx$$

$$\text{L.H.S.} = \int_0^{\infty} e^{-x^2-2bx-b^2} \cdot e^{b^2} dx$$

$$\text{L.H.S.} = \int_0^{\infty} e^{-(x^2+2bx+b^2)} \cdot e^{b^2} dx$$

$$\text{L.H.S.} = e^{b^2} \int_0^{\infty} e^{-(x+b)^2} dx$$

$$\text{put } x+b=u \Rightarrow \boxed{x=u-b}$$



As  $x = u - b$   
 on differentiating  
 $dx = du$

$$\boxed{dx = du}$$

$$u = x + b$$

$$\text{when } x = 0 \implies u = 0 + b = b$$

$$x = \infty \implies u = \infty + b = \infty$$

$$\therefore \text{L.H.S.} = e^{b^2} \int_0^{\infty} e^{-(x+b)^2} dx$$

$$\text{L.H.S.} = e^{b^2} \int_b^{\infty} e^{-u^2} du$$

(Now Adjust  $\frac{2}{\sqrt{\pi}}$ )

$$\text{L.H.S.} = e^{b^2} \cdot \frac{\sqrt{\pi}}{2} \cdot \frac{2}{\sqrt{\pi}} \int_b^{\infty} e^{-u^2} du$$

$$\text{L.H.S.} = \frac{\sqrt{\pi}}{2} e^{b^2} \cdot \text{erfc}(b)$$

$$\text{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-u^2} du$$

$$\text{Also } \text{erf}(x) + \text{erfc}(x) = 1$$

$$\text{L.H.S.} = \frac{\sqrt{\pi}}{2} e^{b^2} (1 - \text{erf}(b))$$

$$\text{L.H.S.} = \text{R.H.S.}$$

$$\text{hence } \int_0^{\infty} e^{-x^2 - 2bx} dx = \frac{\sqrt{\pi}}{2} e^{b^2} (1 - \text{erf}(b))$$